Computational Finance
Lecture 7

Course nr: WI4151

- Heston stochastic volatility model:
  - Properties of the CIR variance process
  - Affine structure and the characteristic function
- Monte Carlo simulation under the Heston model:
  - Simulation of Correlated Brownian motions
  - Euler and exact simulation of the Heston model
  - Characteristic function of the integrated variance: infinitesimal generators, time change, and change of measure

The need to model Vol as a stochastic variable

- The volatility smile can be fitted to by
  - Adding jumps to GBM, or
  - Making volatility stochastic
- Popular stochastic volatility models in banking industry:
  - SABR model

$$dS_t = \sigma_t S_t^{\beta}\, dW_t, \qquad d\sigma_t = \alpha \sigma_t\, dZ_t, \qquad dW_t\, dZ_t = \rho\, dt$$

    - Hagan formula gives a good approximation of the option prices under SABR model
    - However, it is only accurate within a range of T.
  - Heston model
    - Semi-analytical solution exists for pricing options under Heston's model
    - Exact simulation of Heston's model exists

Heston stochastic volatility model

- Heston model:

$$dS_t = rS_t\,dt + \sqrt{\nu_t}\,S_t\,dW_t^S, \qquad d\nu_t = \kappa(\bar{\nu} - \nu_t)\,dt + \gamma\sqrt{\nu_t}\,dW_t^\nu,$$

  where
  - $\bar{\nu}$ is the long-run mean of the volatility;
  - $\kappa > 0$ is the mean reversion rate at which ν_t reverts to the long-run mean $\bar{\nu}$;
  - $\gamma > 0$ is the volatility of the volatility ν_t;
  - the two Brownian motions are correlated with the correlation parameter $\rho_{S,\nu}$, i.e., $dW_t^S\,dW_t^\nu = \rho_{S,\nu}\,dt$.

Affine diffusion processes

- The SDE of an n-dimensional affine diffusion process can be described as

$$d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\,dt + \sigma(\boldsymbol{X}_t)\,d\boldsymbol{W}_t.$$

where

  - Affine condition: $\mu(\boldsymbol{x}) = \boldsymbol{K}_0 + \boldsymbol{K}_1\boldsymbol{x},\ \ \boldsymbol{K}_0 \in \mathbb{R}^n,\ \ \boldsymbol{K}_1 \in \mathbb{R}^{n\times n}$;
  - $\boldsymbol{W}_t$ is n-dimensional Brownian motion;
  - Affine condition: $\left(\sigma(\boldsymbol{x})\sigma(\boldsymbol{x})^\top\right)_{ij} = (\boldsymbol{H}_0)_{ij} + (\boldsymbol{H}_1)_{ij}\boldsymbol{x}$ with $\boldsymbol{H}_0 \in \mathbb{R}^{n\times n},\ \ \boldsymbol{H}_1 \in \mathbb{R}^{n\times n\times n}$;
  - $\lambda(\boldsymbol{x}) = l_0 + \boldsymbol{l}_1 \cdot \boldsymbol{x}$ with $l_0 \in \mathbb{R},\ \ \boldsymbol{l}_1 \in \mathbb{R}^n$;

Fourier transform of affine models (1/3)

- Consider the transform $\psi^{\boldsymbol{X}}(\boldsymbol{u}, \boldsymbol{X}_t, t, T)$ for a complex vector $\boldsymbol{u} \in \mathbb{C}^n$:

$$\psi^{\boldsymbol{X}}(\boldsymbol{u}, \boldsymbol{X}_t, t, T) = \mathbb{E}\left[\exp\left(-\int_t^T R(\boldsymbol{X}_s)\,ds\right) e^{\boldsymbol{u}\cdot\boldsymbol{X}_T} \,\middle|\, \mathcal{F}_t\right].$$

- Moreover, the short rate is also an affine function of $\boldsymbol{X}_t$, i.e.,

$$R(\boldsymbol{X}_t) = \rho_0 + \boldsymbol{\rho}_1 \cdot \boldsymbol{X}_t, \qquad \rho_0 \in \mathbb{R}, \quad \boldsymbol{\rho}_1 \in \mathbb{R}^n.$$

  Note that the dot operation in $\boldsymbol{l}_1 \cdot \boldsymbol{x}$ and $\boldsymbol{\rho}_1 \cdot \boldsymbol{X}_t$ means the inner product of vectors for $n > 1$, or the scalar product for $n = 1$.

Fourier transform of affine models (2/3)

- A key result of affine diffusion processes is that, under certain technical regularity conditions, it holds that

$$\psi^{\boldsymbol{X}}(\boldsymbol{u}, \boldsymbol{x}, t, T) = \mathbb{E}\left[\exp\left(-\int_t^T R(\boldsymbol{X}_s)\, ds\right) e^{\boldsymbol{u}\cdot\boldsymbol{X}_T} \,\middle|\, \mathcal{F}_t\right] = e^{\alpha(\boldsymbol{u},t,T)+\boldsymbol{\beta}(\boldsymbol{u},t,T)\cdot\boldsymbol{x}}.$$

- α and β satisfy the complex-valued ordinary differential equations (ODEs)

$$\begin{aligned}
\dot{\boldsymbol{\beta}} &= \boldsymbol{\rho}_1 - \boldsymbol{K}_1^\top\boldsymbol{\beta} - \frac{1}{2}\boldsymbol{\beta}^\top\boldsymbol{H}_1\boldsymbol{\beta} - \boldsymbol{l}_1\big(\theta(\boldsymbol{\beta}) - 1\big),\\
\dot{\alpha} &= \rho_0 - \boldsymbol{K}_0\cdot\boldsymbol{\beta} - \frac{1}{2}\boldsymbol{\beta}^\top\boldsymbol{H}_0\boldsymbol{\beta} - l_0\big(\theta(\boldsymbol{\beta}) - 1\big).
\end{aligned}$$

- Note that the derivation of the ODEs are done via inserting the affine form of the ch.f. into the PDE that ch.f. satisfies. The PDE is obtained via applying Feynman–Kac theorem.
- with boundary conditions $\boldsymbol{\beta}(\boldsymbol{u}, T, T) = \boldsymbol{u}$ and $\alpha(\boldsymbol{u}, T, T) = 0$.

Fourier transform of affine models (3/3)

- The random jump size is characterized by its characteristics function

$$\theta(\boldsymbol{c}) = \int_{\mathbb{R}^n} \exp(\boldsymbol{c} \cdot \boldsymbol{z})\, d\nu(\boldsymbol{z}).$$

- Note that $\boldsymbol{\beta}(t)^\top \boldsymbol{H}_1 \boldsymbol{\beta}(t)$ denotes an n-dimensional vector with the k-th element being $\sum_{ij} \beta_i(t)\,(\boldsymbol{H}_1)_{ijk}\,\beta_j(t)$.

- The process of ν_t is called a CIR (Cox–Ingersoll–Ross) process. It belongs to the class of AJD.
- Consider the characteristics function (ch.f.) of ν_T conditional on ν_t, i.e.,

$$\psi^{\nu}(u, \nu_t, t, T) = \mathbb{E}(e^{u\nu_T} \mid \nu_t)\,,$$

  and the ODE coefficients are $K_0 = \kappa\bar{\nu}$, $K_1 = -\kappa$, $H_0 = 0$, and $H_1 = \gamma^2$.
- In the previous lecture we have seen that

$$\psi^{\nu}(u, \nu_t, t, T) = e^{\alpha_1(t)+\beta_1(t)\nu_t}$$

  - $\dot{\beta}_1(t) = \kappa\beta_1(t) - \frac{1}{2}\beta_1(t)^2\gamma^2$
  - $\dot{\alpha}_1(t) = -\kappa\bar{\nu}\beta_1(t)$,
  - with the terminal condition $\beta_1(T) = u$ and $\alpha_1(T) = 0$.

The affine property of the stochastic variance (2/4)

- It is easier for expression if we reparametrize

$$\psi^{\nu}(u, \nu_t, t, T) = e^{\alpha(T-t)+\beta(T-t)\nu_t},$$

with

$$-\dot{\beta} = \kappa\beta - \frac{1}{2}\beta^2\gamma^2$$
$$-\dot{\alpha} = -\kappa\bar{\nu}\beta(t),$$

with the initial condition $\beta(0) = u$ and $\alpha(0) = 0$.

- $\dot{\beta} = -\kappa\beta + \frac{1}{2}\beta^2\gamma^2$ is a Ricatti type of ODE. We can solve it analytically.

The affine property of the stochastic variance (3/4)

- Since $\dot{\beta} = -\kappa\beta + \frac{1}{2}\beta^2\gamma^2$,
  - the change of variable $g = 1/\beta$ gives $\dot{g} = \kappa g - \frac{1}{2}\gamma^2$.
  - Further we have $(dg - \kappa g)e^{-\kappa t} = -\frac{1}{2}\gamma^2 e^{-\kappa t}\,dt$, which gives

$$d(ge^{-\kappa t}) = -\frac{1}{2}\gamma^2 e^{-\kappa t}\,dt,$$

  - implying

$$g(t)e^{-\kappa t} - g(0) = -\frac{\gamma^2}{2}\int_0^t e^{-\kappa s}\,ds.$$

- Therefore,
  - $g(t) = e^{\kappa t}\left(g(0) - \dfrac{\gamma^2(1-e^{-\kappa t})}{2\kappa}\right)$, with $g(0) = \dfrac{1}{u}$.
- Thus, $\beta(t) = \dfrac{ue^{-\kappa t}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(t)}$, with $\xi_\kappa(t) = \dfrac{1-e^{-\kappa t}}{\kappa}$.

The affine property of the stochastic variance (4/4)

- $-\dot{\alpha} = -\kappa\bar{\nu}\beta(t)$ leads to

$$\alpha(t) = \alpha(0) + \int_0^t \kappa\bar{\nu}\beta(s)\,ds = \kappa\bar{\nu}\int_0^t \beta(s)\,ds = \kappa\bar{\nu}u\int_0^t \frac{e^{-\kappa s}}{1-\frac{u\gamma^2}{2}\xi_\kappa(s)}\,ds.$$

- Note that the numerator of the integrand, i.e., $e^{-\kappa s}$ is the derivative of $\xi_\kappa(s)$. So we have

$$\alpha(t) = \kappa\bar{\nu}u\int_0^t \frac{e^{-\kappa s}}{1-\frac{u\gamma^2}{2}\xi_\kappa(s)}\,ds = \kappa\bar{\nu}u\cdot\frac{-2}{u\gamma^2}\ln\left(1-\frac{u\gamma^2}{2}\xi_\kappa(t)\right) = -\frac{2\kappa\bar{\nu}}{\gamma^2}\ln\left(1-\frac{u\gamma^2}{2}\xi_\kappa(t)\right).$$

The distribution of the variance at T

- In summary, we have

$$\mathbb{E}(e^{u\nu_T} \mid \nu_t) = \psi^\nu(u, \nu_t, t, T) = \exp\left(-\frac{2\kappa\bar{\nu}}{\gamma^2}\ln\left(1 - \frac{u\gamma^2}{2}\xi_\kappa(T-t)\right) + \frac{ue^{-\kappa(T-t)}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(T-t)}\nu_t\right)$$

$$= \left(1 - \frac{u\gamma^2}{2}\xi_\kappa(T-t)\right)^{-2\kappa\bar{\nu}/\gamma^2}\exp\left(\frac{ue^{-\kappa(T-t)}}{1 - \frac{u\gamma^2}{2}\xi_\kappa(T-t)}\nu_t\right).$$

- Note that our derivation of the solutions to the ODEs assume u is real number. However, the derived form of $\mathbb{E}(e^{u\nu_T} \mid \nu_t)$ is in general correct for a complex u (that's within the convergence domain of $\mathbb{E}(e^{u\nu_T} \mid \nu_t)$ given ν_t).
- The formula above turns out to be identical to the ch.f. of a non-central Chi-squared random variable. Therefore, the distribution of ν_T given ν_t is now known.

Properties of the CIR process

- Conditional on ν_t, the distribution of ν_T is known, i.e.,

$$\nu_T \mid \nu_t \sim \frac{Y}{2c}.$$

  where

  - $c = \dfrac{2\kappa}{\gamma^2(1 - e^{-\kappa(T-t)})}$ is a constant;
  - Y follows a noncentral Chi-squared distribution with $\dfrac{4\kappa\bar{\nu}}{\gamma^2}$ degrees of freedom and non-centrality parameter $2ce^{-\kappa(T-t)}\nu_t$.

- Let $(X_1, X_2, \cdots X_d)$ be d independent, normally distributed random variable with means μ_i and unit variances. Then the random variable $\sum_{i=1}^{k} X_i^2$ is distributed according to the noncentral chi-squared distribution.

- The noncentral chi-squared distribution has two parameters: d is the number of degrees of freedom, and $\lambda = \sum_{i=1}^{k} \mu_i^2$ is the non-centrality parameter. Note that $\lambda > 0$.

Noncentral chi-squared distribution

- The ch.f. of a non-central Chi-squared distribution is

$$\mathbb{E}\left[e^{i\omega\lambda}\right] = \mathbb{E}\left[e^{i\omega\sum_{i=1}^{k}X_i^2}\right] = \frac{1}{(1-2i\omega)^{k/2}}e^{\frac{i\lambda\omega}{1-2i\omega}}$$

- The pdf of the non-central Chi-squared distribution:

$$f(x;k,\lambda) = \frac{1}{2}e^{-(x+\lambda)/2}\left(\frac{x}{\lambda}\right)^{k/4-1/2} I_{\frac{k}{2}-1}(\sqrt{\lambda x})$$

  - where $I_\nu(y)$ is a modified Bessel function of the first kind given by

$$I_\nu(y) = \left(\frac{y}{2}\right)^\nu \sum_{j=0}^{\infty}\frac{\left(\frac{y^2}{4}\right)^j}{j!\Gamma(\nu+j+1)},$$

    with Γ being the Gamma function.

- The cdf $F(x;k,\lambda) = 1 - Q_{\frac{k}{2}}(\sqrt{\lambda},\sqrt{x})$, with the Marcum Q-function $Q_M(a,b)$ defined as

$$Q_M(a,b) = \frac{1}{a^{M-1}}\int_b^{+\infty} x^M \exp\left(-\frac{x^2+a^2}{2}\right) I_{M-1}(ax)\,dx.$$

Paths of the CIR process

- Obviously the CIR process does not allow negative values for ν_t. If the Feller condition holds, i.e., $2\kappa\bar{\nu} \geq \gamma^2$, ν_t stays above zero. Otherwise, ν_t may reach 0 and stay at 0.

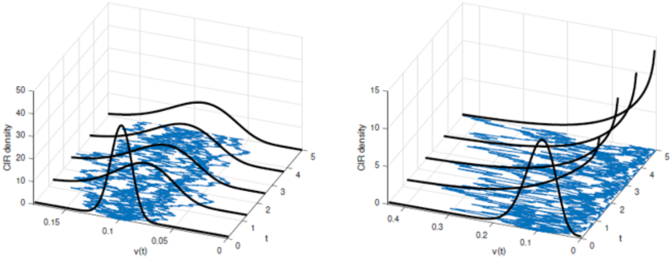


Figure: Paths and the corresponding probability density function (PDF) for the CIR process in the cases where the Feller condition is satisfied (left) and is not satisfied (right). Simulations were performed with $\kappa = 0.5$, $\nu_0 = 0.1$, $\bar{\nu} = 0.1$. Left: $\gamma = 0.1$; Right: $\gamma = 0.35$.

The Feller condition

- The pdf of the non-central Chi-squared distribution

$$f(x;k,\lambda) = \frac{1}{2} e^{-(x+\lambda)/2} \left(\frac{x}{\lambda}\right)^{k/4-1/2} I_{\frac{k}{2}-1}(\sqrt{\lambda x})$$

requires $\frac{k}{2} - 1 \geq 0$ to prevent the pdf blowing up at zero ($x \to 0^+$, $I_\nu(x) \sim (x/2)^\nu / \Gamma(\nu+1)$).

- In the case of ν_t, $k = \dfrac{4\kappa\bar{\nu}}{\gamma^2}$. It requires

$$q_F \stackrel{\text{def}}{=} \frac{2\kappa\bar{\nu}}{\gamma^2} - 1 \geq 0,$$

which is another perspective of the Feller condition.

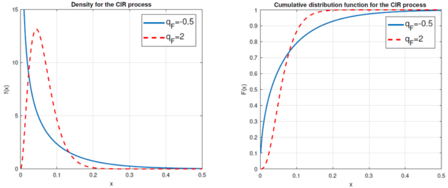


Figure: $q_F = 2$ corresponds to the parameter set for which the Feller condition holds, i.e., $T = 5$, $\kappa = 0.5$, $\nu_0 = 0.2$, $\bar{\nu} = 0.05$ and $\gamma = 0.129$. $q_F = -0.5$ corresponds to the parameter set for which the Feller condition is violated, i.e., $T = 5$, $\kappa = 0.5$, $\nu_0 = 0.2$, $\bar{\nu} = 0.05$ and $\gamma = 0.316$.

Multi-dimensional Brownian motions

- Let

$$\boldsymbol{W}_t = \begin{bmatrix} W_t^{(1)} \\ W_t^{(2)} \\ \vdots \\ W_t^{(n)} \end{bmatrix}$$

  be a vector of correlated Brownian motions.

- For $i \neq j$,

$$\mathbb{E}\left[W_t^{(i)} W_t^{(j)}\right] = \rho_{ij} t, \qquad dW_t^{(i)}\, dW_t^{(j)} = \rho_{ij}\, dt.$$

- The coefficients ρ_{ij} are the instantaneous correlations.
- A standard construction starts from independent Brownian motions and applies a Cholesky factor of the correlation matrix.

Cholesky decomposition (1/3)

- Cholesky decomposition of a real, positive semi-definite symmetric matrix $\mathbf{\Sigma}$ is a decomposition of the form
$$\mathbf{\Sigma} = \boldsymbol{L}\boldsymbol{L}^\top,$$
where $\boldsymbol{L}$ is a lower triangular matrix with real and positive diagonal entries, and $\boldsymbol{L}^\top$ is the transpose of $\boldsymbol{L}$.
- For a given 2×2 correlation matrix $\mathbf{\Sigma}$, the Cholesky decomposition is given as
$$\mathbf{\Sigma} := \begin{bmatrix} 1 & \rho_{1,2} \\ \rho_{1,2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \rho_{1,2} & \sqrt{1-\rho_{1,2}^2} \end{bmatrix} \begin{bmatrix} 1 & \rho_{1,2} \\ 0 & \sqrt{1-\rho_{1,2}^2} \end{bmatrix} = \boldsymbol{L}\boldsymbol{L}^\top.$$

Cholesky decomposition (2/3)

- Given a vector of two independent Brownian motions $\widetilde{W}_t^{(1)}$ and $\widetilde{W}_t^{(2)}$, i.e.

$$\widetilde{\mathbf{W}}_t = \begin{bmatrix} \widetilde{W}_t^{(1)} \\ \widetilde{W}_t^{(2)} \end{bmatrix},$$

- we can generate a vector of two correlated Brownian motions $W_t^{(1)}, W_t^{(2)}$ as follows,

$$\mathbf{W}_t = \mathbf{L}\widetilde{\mathbf{W}}_t = \begin{bmatrix} 1 & 0 \\ \rho_{1,2} & \sqrt{1-\rho_{1,2}^2} \end{bmatrix} \begin{bmatrix} \widetilde{W}_t^{(1)} \\ \widetilde{W}_t^{(2)} \end{bmatrix} = \begin{bmatrix} \widetilde{W}_t^{(1)} \\ \rho_{1,2}\widetilde{W}_t^{(1)} + \sqrt{1-\rho_{1,2}^2}\,\widetilde{W}_t^{(2)} \end{bmatrix}.$$

Cholesky decomposition (3/3)

- We can verify that the correlation of $W_t^{(1)}$ and $W_t^{(2)}$ is indeed $\rho_{1,2}$:

$$\text{Cov}(W_t^{(1)}, W_t^{(2)}) = \mathbb{E}\left[W_t^{(1)} W_t^{(2)}\right] = \rho_{1,2}\mathbb{E}\left[(\widetilde{W}_t^{(1)})^2\right] = \rho_{1,2}t.$$

- Therefore

$$\text{Corr}(W_t^{(1)}, W_t^{(2)}) = \frac{\text{Cov}(W_t^{(1)}, W_t^{(2)})}{\sqrt{\text{Var}(W_t^{(1)})}\sqrt{\text{Var}(W_t^{(2)})}} = \rho_{1,2}.$$

- In general, we can create a vector of correlated Brownian motions using the Cholesky decomposition for any dimensionality.

Correlated Paths

- In the first figure, the two Brownian motions are governed by a negative correlation parameter $\rho_{1,2}$, in the second figure $\rho_{1,2} = 0$, while in the third figure a positive correlation $\rho_{1,2}$ is used.

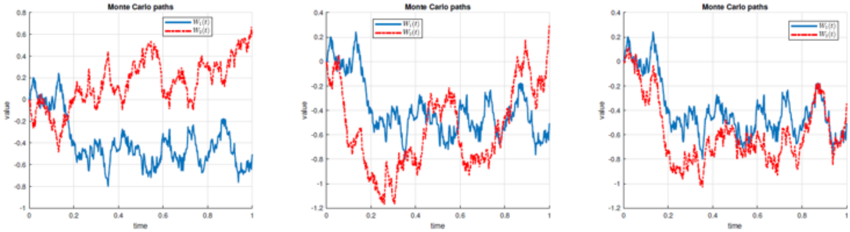


Figure: Monte Carlo Brownian motion $W_i(t)$ paths with different correlations, $\mathbb{E}[W_1(t)W_2(t)] = \rho_{1,2}(t)$; middle: zero correlation; left: negative correlation ($\rho_{1,2} < 0$); right: positive correlation ($\rho_{1,2} > 0$).

Heston stochastic volatility model

- Now we can re-write the Heston model as:

$$d \ln S_t = \left(r - \frac{1}{2}\nu_t\right) dt + \sqrt{\nu_t}\left(\rho_{S,\nu}\, dW_t^S + \sqrt{1-\rho_{S,\nu}^2}\, dW_t^\nu\right),$$

$$d\nu_t = \kappa(\bar{\nu} - \nu_t)dt + \gamma\sqrt{\nu_t}dW_t^\nu,$$

  where W_t^S and W_t^ν are independent Brownian motions.
  (Note that we've applied Cholesky decomposition)
- Also recall that: when the Feller condition, $2\kappa\bar{\nu} > \gamma^2$, is satisfied, ν_t can never reach zero. However, if this condition does not hold, the origin is accessible. In both cases, ν_t cannot become negative.

Interpretation of model parameters (1/4)

- In the Heston model, each parameter has a specific effect on the implied volatility curve.
- To analyze the parameter effects numerically with an example, we use here the following set of reference parameters, $\rho_{S,\nu} = 0,\ \kappa = 1,\ \gamma = 0.1,\ \nu_0 = 0.05$, and $\bar{\nu} = 0.1$.
- We change individual parameters while keeping the other fixed. For each parameter set option prices have been generated and inserted in a Newton–Raphson algorithm to determine the implied volatilities.

Interpretation of model parameters (2/4)

- Impact of correlation $\rho_{S,\nu}$ and the volatility-of-volatility parameter γ.
  - For $\rho_{S,\nu} = 0$, a higher value of γ gives a more profound implied volatility smile. A higher volatility of volatility parameter increases the implied volatility curvature.
  - As the correlation between the stock price and volatility, $\rho_{S,\nu}$, gets increasingly negative, the slop of the skew in the implied volatility curve increases.

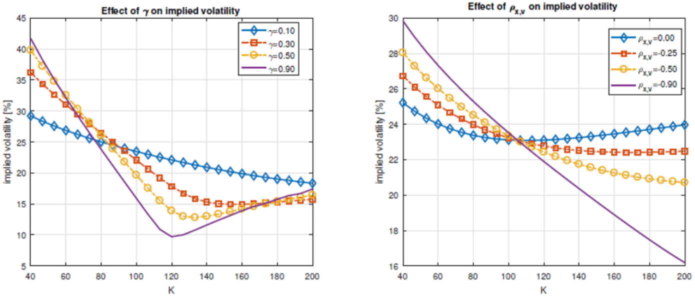


Figure: Impact of variation of the Heston vol-of-vol parameter γ [0.1, 0.3, 0.5, 0.9](left side), and correlation parameter $\rho_{S,\nu}$[0., -0.25, -0.5, -0.9] (right side), on the implied volatility, as a function of strike price K.

Interpretation of model parameters (3/4)

- Speed of mean reversion κ has a limited effect on the implied volatility smile or skew, up to $1\% - 2\%$. κ determines the speed at which the volatility converges to the long-term volatility $\bar{\nu}$, see the graph on the right side, which shows the at-the-money (ATM) implied volatility for different κ.
- With $\bar{\nu} = 10\%$ ($\sqrt{\bar{\nu}} \approx 31.62\%$), a large value of κ implies fast convergence of the implied volatility to $\sqrt{\bar{\nu}}$.

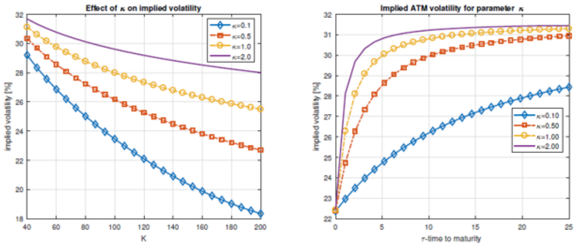


Figure: Impact of variation of the Heston parameter κ on the implied volatility as a function of strike K [0.1, 0.5, 1.0, 2.0] (left side), and impact of variation of κ on the ATM volatility, as a function of $\tau = T - t$ [0.1, 0.5, 1.0, 2.0] (right side).

Interpretation of model parameters (4/4)

- ν_0, the initial variance, and $\bar{\nu}$, the long-term variance level, have a similar effect on the implied volatility curve.
- The effect of these two parameters depends on the value of κ, which controls the speed at which the implied volatility converges from ν_0 to $\bar{\nu}$.

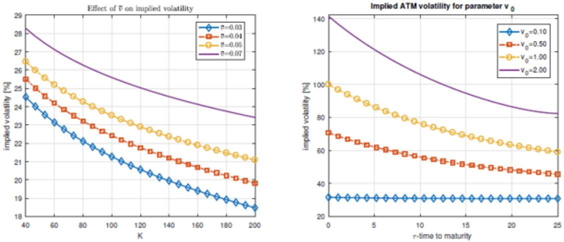


Figure: Impact of changing ν_0 and $\bar{\nu}$ on the Heston implied volatility; left side: $\bar{\nu}$ as a function of the strike K, [0.03, 0.04, 0.05, 0.07]; right side: ν_0 as a function of time to maturity $\tau = T - t$ [0.1, 0.5, 1.0, 2.0].

Black–Scholes vs Heston

- Heston model generates heavier tails of the log returns, compared to Black–Scholes.
- The volatility smile can be better represented by parameters combinations of Heston.
- Set $T = 2$, $\nu_0 = 0.1$, $r = 0.05$, $S_0 = 1$, $\kappa = 0.2$, $\bar{\nu} = 0.3$, $\gamma = 0.1$, and $\rho = -0.8$.

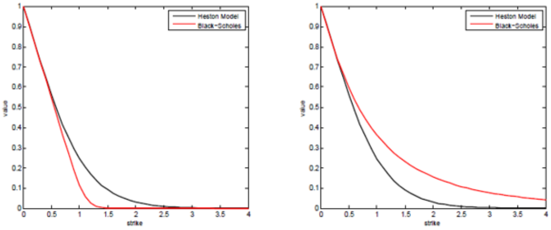


Figure: LEFT: $\sigma^{BS} = \sqrt{\nu_0}$, RIGHT: $\sigma^{BS} = 60\%$

Figure: LEFT: $\sigma^{BS} = \sqrt{\nu_0}$, RIGHT: $\sigma^{BS} = 60\%$.

Recall the definition of affine diffusion models

- The SDE of an n-dimensional affine diffusion process can be described as

$$d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\,dt + \sigma(\boldsymbol{X}_t)\,d\boldsymbol{W}_t$$

  where

  - Affine condition: $\mu(\boldsymbol{x}) = \boldsymbol{K}_0 + \boldsymbol{K}_1\boldsymbol{x},\ \boldsymbol{K}_0 \in \mathbb{R}^n,\ \boldsymbol{K}_1 \in \mathbb{R}^{n\times n}$;
  - $\boldsymbol{W}_t$ is n-dimensional Brownian motion;
  - Affine condition: $\left(\sigma(\boldsymbol{x})\sigma(\boldsymbol{x})^\top\right)_{ij} = (\boldsymbol{H}_0)_{ij} + (\boldsymbol{H}_1)_{ij}\boldsymbol{x}$ with $\boldsymbol{H}_0 \in \mathbb{R}^{n\times n},\ \boldsymbol{H}_1 \in \mathbb{R}^{n\times n\times n}$;
  - $\lambda(\boldsymbol{x}) = l_0 + \boldsymbol{l}_1 \cdot \boldsymbol{x}$ with $l_0 \in \mathbb{R},\ \boldsymbol{l}_1 \in \mathbb{R}^n$;

Is Heston model affine?

- From the definition of the Heston model we have

$$\begin{cases} dS_t = rS_t dt + \sqrt{\nu_t} S_t \, dW_t^S \\ d\nu_t = \kappa(\bar{\nu} - \nu_t)dt + \gamma\sqrt{\nu_t} dW_t^\nu \end{cases}$$

- Check the affine condition on the covariance matrix:

$$\sigma(\boldsymbol{X}_t)\sigma(\boldsymbol{X}_t)^\top = \begin{bmatrix} \nu_t S_t^2 & \gamma\rho_{S,\nu}\nu_t S_t \\ \gamma\rho_{S,\nu}\nu_t S_t & \gamma^2\nu_t \end{bmatrix}.$$

- No, it is not affine!

Restore the affine condition (1/2)

- Moving to the log-domain, i.e. $X_t = \ln S_t$, we have

$$\begin{cases} dX_t = \left(r - \frac{1}{2}\nu_t\right) dt + \sqrt{\nu_t} dW_t^S \\ d\nu_t = \kappa(\bar{\nu} - \nu_t)dt + \gamma\sqrt{\nu_t} dW_t^\nu \end{cases}$$

- In matrix-vector-product form,

$$\begin{bmatrix} d\ln S_t \\ d\nu_t \end{bmatrix} = \left(\underbrace{\begin{bmatrix} r \\ \kappa\bar{\nu} \end{bmatrix}}_{\boldsymbol{K}_0} + \underbrace{\begin{bmatrix} 0 & -1/2 \\ 0 & -\kappa \end{bmatrix}}_{\boldsymbol{K}_1} \begin{bmatrix} \ln S_t \\ \nu_t \end{bmatrix} \right) dt + \sqrt{\nu_t} \begin{bmatrix} 1 & 0 \\ \gamma\rho_{S,\nu} & \gamma\sqrt{1-\rho_{S,\nu}^2} \end{bmatrix} \begin{bmatrix} d\widetilde{W}_1(t) \\ d\widetilde{W}_2(t) \end{bmatrix},$$

  - Affine in the drift term

Restore the affine condition (2/2)

- The covariance matrix

$$\sigma\sigma^\top = \sqrt{\nu_t}\begin{bmatrix} 1 & 0 \\ \gamma\rho_{S,\nu} & \gamma\sqrt{1-\rho_{S,\nu}^2} \end{bmatrix} \sqrt{\nu_t}\begin{bmatrix} 1 & \gamma\rho_{S,\nu} \\ 0 & \gamma\sqrt{1-\rho_{S,\nu}^2} \end{bmatrix}$$
$$= \nu_t \begin{bmatrix} 1 & \gamma\rho_{S,\nu} \\ \gamma\rho_{S,\nu} & \gamma^2 \end{bmatrix}$$

  - The covariance matrix is also affine!

- Match to our template: $\left(\sigma(\boldsymbol{x})\sigma(\boldsymbol{x})^\top\right)_{ij} = (\boldsymbol{H}_0)_{ij} + (\boldsymbol{H}_1)_{ij}\boldsymbol{x}$ with $\boldsymbol{H}_0 \in \mathbb{R}^{n\times n}$, $\boldsymbol{H}_1 \in \mathbb{R}^{n\times n\times n}$, we can see that
  - $\boldsymbol{H}_0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
  - $\boldsymbol{H}_1$ is a $2\times 2\times 2$ tensor (i.e., two 2×2 matrices), of which the first element of $\boldsymbol{H}_1$ is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, and the second is $\begin{bmatrix} 1 & \gamma\rho_{S,\nu} \\ \gamma\rho_{S,\nu} & \gamma^2 \end{bmatrix}$.

Recall the key result of affine diffusion processes

- For affine diffusion processes, it holds that

$$\psi^{\boldsymbol{X}}(\boldsymbol{u}, \boldsymbol{x}, t, T) = \mathbb{E}\left[\exp\left(-\int_t^T R(\boldsymbol{X}_s)\, ds\right) e^{\boldsymbol{u}\cdot\boldsymbol{X}_T} \,\middle|\, \mathcal{F}_t\right] = e^{\alpha(\boldsymbol{u},t,T)+\boldsymbol{\beta}(\boldsymbol{u},t,T)\cdot\boldsymbol{x}}.$$

  - α and β satisfy the complex-valued ordinary differential equations (ODEs)

$$\dot{\boldsymbol{\beta}} = \boldsymbol{\rho}_1 - \boldsymbol{K}_1^\top \boldsymbol{\beta} - \frac{1}{2}\boldsymbol{\beta}^\top \boldsymbol{H}_1 \boldsymbol{\beta},$$
$$\dot{\alpha} = \rho_0 - \boldsymbol{K}_0 \cdot \boldsymbol{\beta} - \frac{1}{2}\boldsymbol{\beta}^\top \boldsymbol{H}_0 \boldsymbol{\beta}$$

  - with boundary conditions $\boldsymbol{\beta}(\boldsymbol{u}, T, T) = \boldsymbol{u}$ and $\alpha(\boldsymbol{u}, T, T) = 0$.

Characteristics function of the Heston model

- With $u = i\omega$ being a pure imaginary number, the ch.f. of $y_t \stackrel{\text{def}}{=} \ln S_t$ can be found by solving the ODEs

$$\mathbb{E}[e^{uy_T} \mid y_t, \nu_t] = \psi^{\mathbf{Y}}(u, \nu_t, t, T) = e^{\alpha(T-t,u)+uy_t+\beta(T-t,u)\nu_t},$$

where, letting $b = \gamma\rho_{S,\nu}u - \kappa$, $a = u(1-u)$, and $s = \sqrt{b^2 + a\gamma^2}$, we have

$$\beta(\tau, u) = \frac{a(1-e^{-s\tau})}{2s - (s+b)(1-e^{-s\tau})},$$

$$\alpha(\tau, u) = -r\tau + ru\tau - \kappa\bar{\nu}\left(\frac{s+b}{\gamma^2}\tau + \frac{2}{\gamma^2}\ln\left[1 - \frac{s+b}{2s}(1-e^{-s\tau})\right]\right).$$

References

- Darrell Duffie, Jun Pan, and Kenneth Singleton, *Transform Analysis and Asset Pricing for Affine Jump-Diffusions*, Econometrica, 68(6), 2000.
  https://doi.org/10.1111/1468-0262.00164

Multidimensional diffusion processes (1/2)

- First, consider $\boldsymbol{W}_t = \boldsymbol{L}\widetilde{\boldsymbol{W}}_t$, where $\widetilde{\boldsymbol{W}}_t$ is an n-dimensional vector of independent Brownian motions and $\boldsymbol{L}$ is the Cholesky decomposition of an $n \times n$ covariance matrix $\boldsymbol{C}$.
- Note that
$$d\widetilde{\boldsymbol{W}}_t \, d\widetilde{\boldsymbol{W}}_t^\top = dt \cdot \boldsymbol{I}_{n\times n},$$
where $\boldsymbol{I}_{n\times n}$ is an identity matrix.
- Therefore,
$$d\boldsymbol{W}_t \, d\boldsymbol{W}_t^\top = \boldsymbol{L} \, d\widetilde{\boldsymbol{W}}_t \, d\widetilde{\boldsymbol{W}}_t^\top \boldsymbol{L}^\top = \boldsymbol{L}\boldsymbol{I}_{n\times n}\boldsymbol{L}^\top \, dt = \boldsymbol{C} \, dt.$$

Multidimensional diffusion processes (2/2)

- E.g., an n-dimensional diffusion process is written as

$$d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\,dt + \sigma(\boldsymbol{X}_t)\,d\boldsymbol{W}_t.$$

$$d\begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} dt + \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix} d\begin{bmatrix} W_1 \\ \vdots \\ W_n \end{bmatrix}.$$

where $\boldsymbol{W}_t$ is an n-dimensional correlated Brownian motion.

- Note that $d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\,dt + \sigma(\boldsymbol{X}_t)\,d\boldsymbol{W}_t$ is equivalent to

$$d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\,dt + \sigma(\boldsymbol{X}_t)\,\boldsymbol{L}\,d\widetilde{\boldsymbol{W}}_t.$$

Ito's lemma for a multidimensional diffusion process (1/2)

- Consider $g(t, \boldsymbol{x})$ which is a function from $\mathbb{R} \times \mathbb{R}^n$ to $\mathbb{R}$, and

$$\boldsymbol{X}_t = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

which is an n-dimensional diffusion process, i.e.,

$$d\boldsymbol{X}_t = \mu(\boldsymbol{X}_t)\, dt + \sigma(\boldsymbol{X}_t)\, d\boldsymbol{W}_t.$$

- Suppose the first order derivative of g w.r.t. t and the second order derivative of g w.r.t. x_i exist, Ito's lemma is as follows

$$dg(t, \boldsymbol{X}_t) = \frac{\partial g}{\partial t}\, dt + \sum_{i=1}^{n} \frac{\partial g}{\partial x_i}\, dx_i + \frac{1}{2} \sum_{1 \le i,j \le n} \frac{\partial^2 g}{\partial x_i \partial x_j}\, dx_i\, dx_j.$$

Ito's lemma for a multidimensional diffusion process (2/2)

- In Matrix-Vector expressions, we can write

$$
\begin{aligned}
dg(t, \boldsymbol{X}_t) &= \frac{\partial g}{\partial t}\, dt + (\nabla_{\boldsymbol{X}} g)^\top d\boldsymbol{X}_t + \frac{1}{2}(d\boldsymbol{X}_t)^\top (H_{\boldsymbol{X}} g)\, d\boldsymbol{X}_t \\
&= \left(\frac{\partial g}{\partial t} + (\nabla_{\boldsymbol{X}} g)^\top \mu(\boldsymbol{X}_t) + \frac{1}{2}\,\mathrm{Tr}\Big(\sigma(\boldsymbol{X}_t)^\top (H_{\boldsymbol{X}} g)\, \sigma(\boldsymbol{X}_t)\Big)\right) dt + (\nabla_{\boldsymbol{X}} g)^\top \sigma(\boldsymbol{X}_t)\, d\boldsymbol{W}_t,
\end{aligned}
$$

- $\mathrm{Tr}(\cdot)$ is the trace of a matrix (sum of the elements on the diagonal),
- $\nabla_{\boldsymbol{X}} g = \left[\frac{\partial g}{\partial x_1}, \cdots, \frac{\partial g}{\partial x_n}\right]^\top$ is the gradient of g w.r.t. $\boldsymbol{X}$, and
- $H_{\boldsymbol{X}} g$ is the Hessian matrix of g w.r.t. $\boldsymbol{X}$, i.e.,

$$
H_{\boldsymbol{X}} g = \begin{bmatrix}
\frac{\partial^2 g}{\partial x_1^2} & \frac{\partial^2 g}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 g}{\partial x_1 \partial x_n} \\
\frac{\partial^2 g}{\partial x_2 \partial x_1} & \frac{\partial^2 g}{\partial x_2^2} & \cdots & \frac{\partial^2 g}{\partial x_2 \partial x_n} \\
\vdots & \vdots & \ddots & \vdots \\
\frac{\partial^2 g}{\partial x_n \partial x_1} & \frac{\partial^2 g}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 g}{\partial x_n^2}
\end{bmatrix}.
$$

PDE under the Heston model

- Again, using Ito's lemma we can find the PDE of the price of a contingent claim V_t:

$$
\begin{aligned}
d\left(e^{-rt}V_t\right) &= e^{-rt}\left(-rV_t\,dt + dV_t\right)\\
&= e^{-rt}\Bigg(-rV_t\,dt + \frac{\partial V}{\partial t}\,dt + \frac{\partial V}{\partial S}\,dS + \frac{\partial V}{\partial \nu}\,d\nu + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\,dS\,dS\\
&\qquad + \frac{1}{2}\frac{\partial^2 V}{\partial \nu^2}\,d\nu\,d\nu + \frac{\partial^2 V}{\partial S\,\partial \nu}\,dS\,d\nu\Bigg)\\
&= e^{-rt}\Bigg(-rV_t + \frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \kappa(\bar{\nu}-\nu_t)\frac{\partial V}{\partial \nu} + \frac{1}{2}\nu_t S^2\frac{\partial^2 V}{\partial S^2}\\
&\qquad + \frac{1}{2}\gamma^2\nu_t\frac{\partial^2 V}{\partial \nu^2} + \gamma\nu_t S\rho_{S,\nu}\frac{\partial^2 V}{\partial S\,\partial \nu}\Bigg)dt\\
&\quad + e^{-rt}\left(\sqrt{\nu_t}S\frac{\partial V}{\partial S}\,dW_t^S + \gamma\sqrt{\nu_t}\frac{\partial V}{\partial \nu}\,dW_t^\nu\right)
\end{aligned}
$$

where we used $dS\,d\nu = \gamma\nu_t S\rho_{S,\nu}\,dt$.

- Thus, for $e^{-rt}V_t$ being a martingale, the following PDE needs to be satisfied

$$
-rV_t + \frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \kappa(\bar{\nu}-\nu_t)\frac{\partial V}{\partial \nu} + \frac{1}{2}\nu_t S^2\frac{\partial^2 V}{\partial S^2} + \frac{1}{2}\gamma^2\nu_t\frac{\partial^2 V}{\partial \nu^2} + \gamma\nu_t S\rho_{S,\nu}\frac{\partial^2 V}{\partial S\,\partial \nu} = 0.
$$

Heston stochastic volatility model

- Recall the Heston stochastic volatility model under the risk-neutral measure can be represented as:

$$d \ln S_t = \left(r - \frac{1}{2}\nu_t\right) dt + \sqrt{\nu_t}\left(\rho_{S,\nu}\, dW_t^S + \sqrt{1-\rho_{S,\nu}^2}\, dW_t^\nu\right),$$

$$d\nu_t = \kappa(\bar{\nu} - \nu_t)\, dt + \gamma\sqrt{\nu_t}\, dW_t^\nu,$$

where W_t^S and W_t^ν are independent Brownian motions. Note that we've applied Cholesky decomposition.

- Also recall that when the Feller condition $2\kappa\bar{\nu} > \gamma^2$ is satisfied, ν_t can never reach zero. However, if this condition does not hold, the origin is accessible. In both cases, ν_t cannot become negative.

Euler simulation of Heston model

- The Euler scheme of the Heston stochastic volatility is straightforward:

$$\begin{cases} \nu_{t_{i+1}} = \nu_{t_i} + \kappa(\bar{\nu} - \nu_{t_i})(t_{i+1} - t_i) + \gamma\sqrt{\nu_{t_i}}\sqrt{t_{i+1} - t_i} \cdot Z, \\ \ln S_{t_{i+1}} = \ln S_{t_i} + \left(r - \frac{1}{2}\nu_{t_i}\right)(t_{i+1} - t_i) + \sqrt{\nu_{t_i}}\sqrt{t_{i+1} - t_i} \cdot Y. \end{cases}$$

- where $Y = \rho_{S,\nu} \cdot Z + \sqrt{1 - \rho_{S,\nu}^2} \cdot X$, with $X, Z \sim N(0,1)$ and $X \perp Z$.
- $Z \sim N(0,1)$ implies $\mathbb{P}\left(\nu_{t_{i+1}} < 0\right) > 0$.
- Problem: Euler scheme (and any other simulation scheme using naive time discretization) cannot guarantee the simulated volatility to stay above 0.

A quick solution subject to errors

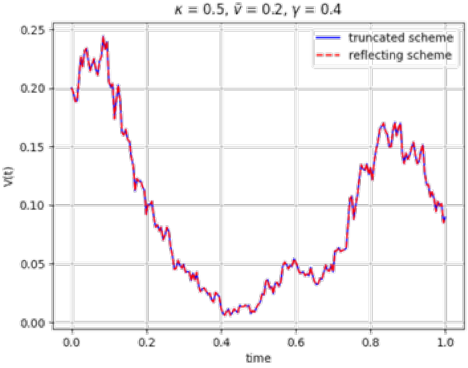
- Before the paper of Broadie and Kaya 2000, a quick solution used in practice is to apply a floor, i.e.

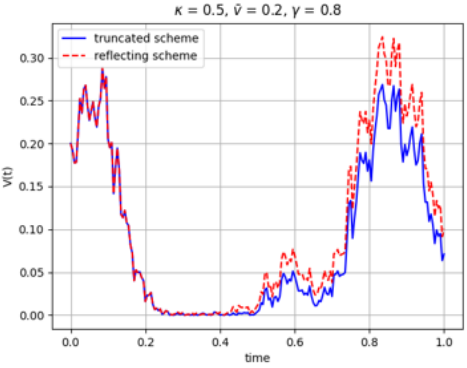
$$\nu_{t_{i+1}} = \max\left(\nu_{t_{i+1}}, \varepsilon\right)$$

  for a tiny positive number ε.
- Problem: flooring at zero is subject to truncation errors, further reducing the efficiency and accuracy of the Euler scheme.
- Another solution that is useful if the Feller condition is not satisfied is the reflection scheme, i.e.

$$\nu_{t_{i+1}} = \left|\nu_{t_{i+1}}\right|$$

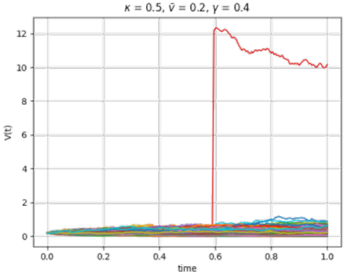
Truncation vs Reflection scheme





How about change of variables?

- Suppose we would try to apply a change of variable $Y_t \equiv \ln(V_t)$ to solve the negative-value issue. We will use a straightforward implementation of the Euler scheme of the SDE for Y_t and then transform back $V_t = \exp(Y_t)$.
- Will this work? NO. Since V can explode in this case.



Exact simulation of the CIR volatility process (1/2)

- A more accurate approach to avoid the negative volatility is to use the property of the CIR process, i.e.,

$$\nu_{t_{i+1}} \mid \nu_{t_i} \sim \frac{Y}{2c}$$

where

  - $c = \dfrac{2\kappa}{\gamma^2\left(1 - e^{-\kappa(t_{i+1}-t_i)}\right)}$ is a constant;
  - $Y \sim \chi^{\star}_{d}(\lambda)$ follows a noncentral Chi-squared distribution with $d = \dfrac{4\kappa\bar{\nu}}{\gamma^2}$ degrees of freedom and non-centrality parameter $\lambda = 2ce^{-\kappa(t_{i+1}-t_i)}\nu_{t_i}$.

- That is, Y as a noncentral Chi-squared random variable can be simulated directly.

Exact simulation of the CIR volatility process (2/2)

$\chi_d^\star(\lambda)$ has the following tractable properties:

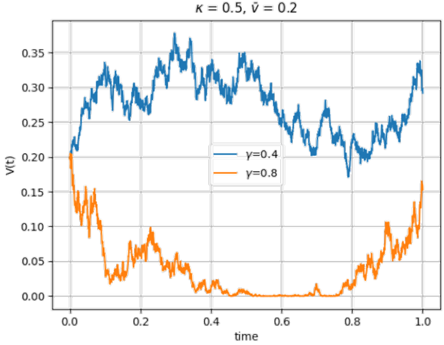
- If $d > 1$, $(\chi_d^\star(\lambda))^2 = (\chi_1^\star(\lambda))^2 + \chi_{d-1}^2$, where χ_{d-1}^2 is a Chi-squared random variable with $d-1$ degrees of freedom.
  - $(\chi_1^\star(\lambda))^2$ can be simulated using the identity probability law $(\chi_1^\star(\lambda))^2 \sim \left(Z + \sqrt{\lambda/2}\right)^2$, where $Z \sim N(0,1)$;
  - χ_{d-1}^2 itself can be simulated directly (available in many statistics software packages).
- In the general case, for any $d > 0$, if $J \sim \mathrm{Poisson}(\lambda/2)$, then $(\chi_d^\star(\lambda))^2 \sim \chi_{d+2J}^2$. That is:
  - we can first simulate the Poisson random variable J;
  - then conditional on simulated J, simulate χ_{d+2J}^2.

Important remark on exact simulation of the CIR model

- The non-central chi-squared distribution is not well-defined whenever the Feller condition is not satisfied.
- Note that the Feller condition is only a sufficient condition for the well-definedness of the noncentral chi-squared distribution. There may be cases where the Feller condition is not satisfied but the distribution is still well-defined, but such cases are rare and not of practical significance.
- Solution: Use a transformation to the Log-variance process $\ln(\nu_t)$, then sample from this process and transform back the result. For specifics I refer to Fang and Oosterlee (2011).

Results

- From exact simulation of the CIR volatility process in python:



Exact simulation of the Heston model (1/2)

- Now we know how to exactly simulate the volatility process, by using the property that $\nu_{t_{i+1}} \mid \nu_{t_i}$ is noncentral chi-squared.
- With the help of the Girsanov theorem and time change, we can in fact exactly simulate the Heston model without any discretization or truncation error.
- Notice that

$$S_t = S_u \exp\left(r(t-u) - \frac{1}{2}\int_u^t \nu_s\, ds + \rho_{S,\nu}\int_u^t \sqrt{\nu_s}\, dW_s^\nu + \sqrt{1-\rho_{S,\nu}^2}\int_u^t \sqrt{\nu_s}\, dW_s^S \right),$$

$$\nu_t = \nu_u + \kappa\bar{\nu}(t-u) - \kappa\int_u^t \nu_s\, ds + \gamma\int_u^t \sqrt{\nu_s}\, dW_s^\nu.$$

- Thus, S_t can be simulated exactly if we know how to simulate $\int_u^t \nu_s\, ds$, $\int_u^t \sqrt{\nu_s}\, dW_s^S$, and $\int_u^t \sqrt{\nu_s}\, dW_s^\nu$.

Exact simulation of the Heston model (2/2)

The exact simulation algorithm of the Heston model developed by Broadie and Kaya 2000 is as follows:

- Step 1. Sample from the noncentral chi-squared distribution of ν_t given ν_u.
- Step 2. Sample from the distribution of $\int_u^t \nu_s\, ds$ given ν_t and ν_u.
- Step 3. Generate a sample of $\int_u^t \sqrt{\nu_s}\, dW_s^\nu$ given $\int_u^t \nu_s\, ds$, ν_t, and ν_u:
$$\int_u^t \sqrt{\nu_s}\, dW_s^\nu = \frac{1}{\gamma}\left(\nu_t - \nu_u - \kappa\bar{\nu}(t-u) + \kappa \int_u^t \nu_s\, ds\right).$$
- Step 4. Generate a sample of $\int_u^t \sqrt{\nu_s}\, dW_s^S$ given $\int_u^t \nu_s\, ds$, ν_t, and ν_u:
$$\int_u^t \sqrt{\nu_s}\, dW_s^S \mid \int_u^t \nu_s\, ds \sim N\left(0, \int_u^t \nu_s\, ds\right).$$
- So the difficult part is Step 2.

How to solve step 2?

- For many diffusion processes X_t, the formula of the general Laplace transform of $\mathbb{E}\left(e^{c\int_u^t X_s\,ds} \mid X_u, X_t\right)$ exists analytically for a complex number c.
- Thus the characteristic function (ch.f.) of $\int_u^t X_s\,ds$ is known, and we can solve the CDF of $\int_u^t X_s\,ds$ using the Fourier inversion method,
  - The original paper applied Trapezoidal rule on the inverse Fourier transform
  - Now we can use the COS method.
- Given the CDF function F, we can simulate the random sample using the inverse sampling theorem.
  - That is, if the continuous random variable R has the CDF $F_R(\cdot)$, then R can be simulated as $F_R^{-1}(U)$ with $U \sim \mathrm{Uniform}(0,1)$ and F_R^{-1} is the inverse of $F_R(\cdot)$. To verify it, note

$$\mathbb{P}\left(F_R^{-1}(U) \le r\right) = \mathbb{P}(U \le F_R(r)) = F_R(r).$$

Ch.f. of $\int_u^t \nu_s\, ds$ given ν_t and ν_u

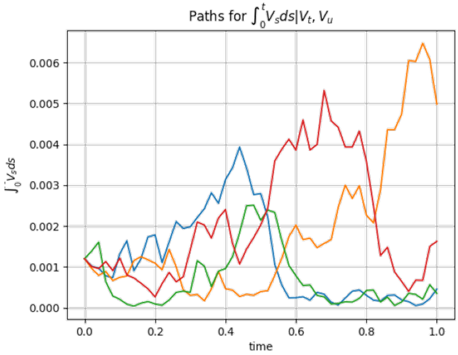
- $\varphi(\omega)$, i.e., the ch.f. of $\int_u^t \nu_s\, ds$ given ν_t and ν_u is then known by plugging $a = -i\omega$ into the Laplace transform, i.e.,

$$
\begin{aligned}
\varphi(\omega) &= \mathbb{E}^Q\left[\exp\left\{i\omega\int_u^t \nu_s\, ds\right\}\Big|\, \nu_u, \nu_t\right]\\
&= \frac{R(\omega)\cdot e^{-\frac{1}{2}(R(\omega)-\kappa)(t-u)}\cdot\left(1-e^{-\kappa(t-u)}\right)}{\kappa\left(1-e^{-R(\omega)(t-u)}\right)}\\
&\quad\times\exp\left\{\frac{\nu_u+\nu_t}{\gamma^2}\left[\frac{\kappa\left(1+e^{-\kappa(t-u)}\right)}{1-e^{-\kappa(t-u)}}-\frac{R(\omega)\left(1+e^{-R(\omega)(t-u)}\right)}{1-e^{-R(\omega)(t-u)}}\right]\right\}\\
&\quad\times\frac{I_{0.5d-1}\left[\sqrt{\nu_t\nu_u}\cdot\frac{4R(\omega)e^{-0.5R(\omega)(t-u)}}{\gamma^2\left(1-e^{-R(\omega)(t-u)}\right)}\right]}{I_{0.5d-1}\left[\sqrt{\nu_t\nu_u}\cdot\frac{4\kappa e^{-0.5\kappa(t-u)}}{\gamma^2\left(1-e^{-\kappa(t-u)}\right)}\right]},
\end{aligned}
$$

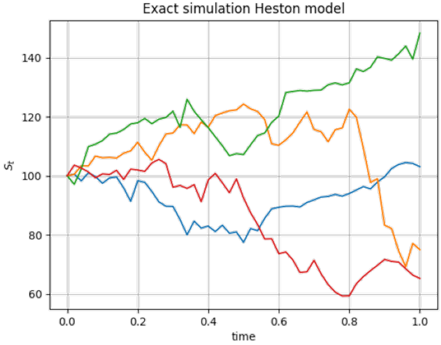
with $R(\omega) = \sqrt{\kappa^2 - 2\gamma^2 i\omega}$, $d = \frac{4\kappa\bar{\nu}}{\gamma^2}$, and $I_y(\cdot)$ is the modified Bessel process of the first kind.

Results

- Paths generated by sampling from $\int_u^t \nu_s \, ds$ given ν_t and ν_u.



- Paths generated using the exact simulation algorithm of the Heston model developed by Broadie and Kaya 2000.



Simulation results of Broadie and Kaya (1/2)

Table 1. Simulation results under the SV model for a European call option.

(a) Simulation with the exact method

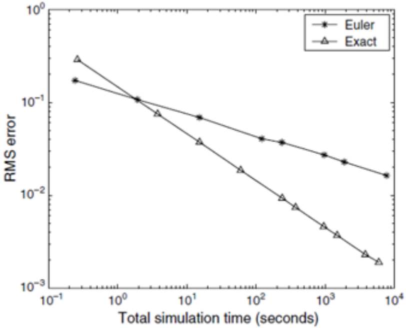
No. of simulation trials	RMS error	Computing time (sec.)
10,000	0.0750	3.8
40,000	0.0373	15.2
160,000	0.0186	60.0
640,000	0.0093	239.4
2,560,000	0.0046	955.7
10,240,000	0.0023	3,822.6

(b) Simulation with the Euler discretization

No. of simulation trials	No. of time steps	Bias	Standard error	RMS error	Computing time (sec.)
10,000	100	0.1543	0.0772	0.1725	0.2
40,000	200	0.1003	0.0381	0.1073	1.9
160,000	400	0.0662	0.0189	0.0689	15.2
640,000	800	0.0395	0.0094	0.0406	121.3
2,560,000	1,600	0.0267	0.0047	0.0272	970.0
10,240,000	3,200	0.0161	0.0023	0.0163	7,758.6

Note. Option parameters: $S = 100$, $K = 100$, $V_0 = 0.010201$, $\kappa = 6.21$, $\theta = 0.019$, $\sigma_v = 0.61$, $\rho = -0.70$, $r = 3.19\%$, $T = 1.0$ year, true option price $= 6.8061$.

Figure 1. Convergence of the RMS errors of the two methods for the option in Table 1 (SV model).



Simulation results of Broadie and Kaya (2/2)

Table 2. Simulation results under the SV model for a European call option.

(a) Simulation with the exact method

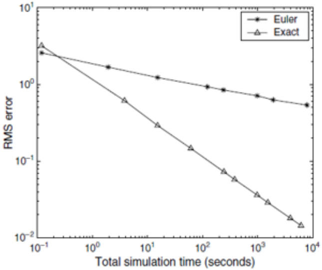
No. of simulation trials	RMS error	Computing time (sec.)
10,000	0.6125	3.8
40,000	0.2904	15.3
160,000	0.1464	61.3
640,000	0.0726	244.5
2,560,000	0.0362	978.6
10,240,000	0.0181	3,916.5

(b) Simulation with the Euler discretization

No. of simulation trials	No. of time steps	Bias	Standard error	RMS error	Computing time (sec.)
10,000	100	2.1962	0.6568	2.2923	0.2
40,000	200	1.6413	0.3247	1.6731	1.9
160,000	400	1.2151	0.1544	1.2248	15.4
640,000	800	0.9265	0.0761	0.9296	122.6
2,560,000	1,600	0.7093	0.0379	0.7103	980.4
10,240,000	3,200	0.5367	0.0188	0.5370	7,838.4

Note. Option parameters: $S = 100$, $K = 100$, $V_0 = 0.09$, $\kappa = 2.00$, $\theta = 0.09$, $\sigma_v = 1.00$, $\rho = -0.30$, $r = 5.00\%$, $T = 5.0$ years, true option price $= 34.9998$.

Figure 2. Convergence of the RMS errors of the two methods for the option in Table 2 (SV model).



(Dis)advantages of the Exact simulation

- Advantages:
  - No discretization & truncation error
  - Higher accuracy (result of the previous)
  - Ability to sample from time t directly (no time steps needed)
- Disadvantages:
  - Algorithm is slow
  - Complicated to implement

The ch.f. of the time-integration of the variance

Derivation of the ch.f. of $\int_u^t \nu_s \, ds$ given ν_t and ν_u relies on the following tools:

- Infinitesimal Generator
- Time change of Brownian motion
- Girsanov's theorem (change of measure)

Infinitesimal generator (1/3)

- Infinitesimal generator encodes a great deal of information about the stochastic process. Pitman and Yor (1982) derives many properties of a class of diffusion processes characterized by some certain form of the infinitesimal generator.
- Consider a continuous time process $X_t \in \mathbb{R}$ following the SDE,

$$dX_t = \mu(X_t)\,dt + \sigma(X_t)\,dW_t,$$

- The infinitesimal generator A of X_t is defined as

$$Af(x) = \lim_{t \to 0} \frac{\mathbb{E}[f(X_t) \mid X_0 = x] - f(x)}{t}.$$

where the set of functions $f : \mathbb{R} \to \mathbb{R}$ such that the limit exists at x is denoted by $D_A(x)$, while D_A denotes the set of functions for which the limit exists for all $x \in \mathbb{R}$.

Infinitesimal generator (2/3)

- There exists a nice connection between the operator A and the SDE of X_t, i.e.,

$$Af(x) = \mu(x)f'(x) + \frac{1}{2}\sigma^2(x)f''(x).$$

E.g., the generator of the CIR volatility process is given by

$$Af(x) = \kappa(\bar{v} - x)f'(x) + \frac{1}{2}\gamma^2 x f''(x).$$

Infinitesimal generator (3/3)

- Pitman and Yor (1982) give the following formula for the n-dimensional squared Bessel process $X_t \in \mathbb{R}$ which has the infinitesimal generator

$$Af(x) = nf'(x) + 2xf''(x).$$

$$\mathbb{E}\left[\exp\left(-\frac{b^2}{2}\int_0^t X_s\,ds\right)\,\middle|\,X_0 = x,\ X_t = y\right] = \frac{bt}{\sinh(bt)}\exp\left(\frac{x+y}{2t}(1 - bt\coth(bt))\right)\frac{I_k\left(\dfrac{b\sqrt{xy}}{\sinh(bt)}\right)}{I_k\left(\dfrac{\sqrt{xy}}{t}\right)}.$$

where b is a complex number and $k = \dfrac{n}{2} - 1$, $I_k(\cdot)$ is the modified Bessel function of the first kind.

- Note that the n-dimensional squared Bessel process X_t (as commonly named in the literature) is characterized by the SDE

$$dX_t = n\,dt + 2\sqrt{X_t}\,dW_t.$$

And X_t is a one-dimensional stochastic process (do not get confused by the name).

Time change of Brownian motion

- We also need the basic formula for time change. By the scaling property of a Brownian motion W_t, $c \cdot W_t \sim W_{c^2 \cdot t}$. This motivates to define the simplest change of time, i.e., a new time clock s

$$s = \eta(t) = \int_0^t c^2 \, du = c^2 t.$$

- It is straightforward to verify that, given $M_t = cW_t$,

$$W_s = M_{\eta^{-1}(s)},$$

  is another Brownian motion at the new time scale s.
- This idea can be generalized (see e.g., Øksendal chapter 8). For a diffusion process defined as

$$dX_t = \mu(X_t) \, dt + \sigma(X_t) \, dW_t,$$

  under the time change $s = \eta(t) = c^2 t$, the transformed process $\widetilde{X}_s := X_{\eta^{-1}(s)}$ satisfies the SDE

$$d\widetilde{X}_s = \frac{\mu(X_t)}{c^2} \, ds + \frac{\sigma(X_t)}{c} \, d\widetilde{W}_s,$$

  where $\widetilde{W}_s$ is another Brownian motion at the new time scale s.

Ideas

- We observe that the generator of the squared Bessel process, i.e., $n \cdot \frac{\partial f}{\partial x}(x) + 2x \frac{\partial^2 f}{\partial x^2}(x)$ and that of the CIR process $\kappa(\bar{\nu} - x) \cdot \frac{\partial f}{\partial x}(x) + \frac{1}{2}\gamma^2 x \cdot \frac{\partial^2 f}{\partial x^2}(x)$ are similar.
- The differences in the second order coefficients of $\frac{\partial^2 f}{\partial x^2}(x)$, i.e., $2x$ versus $\frac{1}{2}\gamma^2 x$ can be addressed by time change of Brownian motion.
- The differences in the first order coefficients of $\frac{\partial f}{\partial x}(x)$, i.e., n versus $\kappa(\bar{\nu} - x)$ can be addressed by the change of measure via the Girsanov theorem.

Time change of the CIR process

- Recall that the CIR volatility process in the risk-neutral measure Q is given as

$$d\nu_t = \kappa(\bar{\nu} - \nu_t)\, dt + \gamma\sqrt{\nu_t}\, dW_t^{\nu}.$$

- By the time change $s = \eta(t) = \frac{\gamma^2}{4}t$,, the time changed process $\rho_s := \nu_{\eta^{-1}(s)}$ satisfies the SDE

$$d\rho_s = \frac{4\kappa}{\gamma^2}(\bar{\nu} - \rho_s)\, ds + 2\sqrt{\rho_s}\, d\widetilde{W}_s^{\nu}.$$

- Let's denote this probability measure by P. Note we have successfully transformed the volatility process such that the second-order coefficient of the infinitesimal generator is the same as that of a squared Bessel process.
- For a complex number a, since $\nu_s = \rho_{\gamma^2 s/4}$

$$\begin{aligned}\mathbb{E}^Q\left[\exp\left(-a\int_u^t \nu_s\, ds\right)\middle|\, \nu_u, \nu_t\right] &= \mathbb{E}^P\left[\exp\left(-a\int_u^t \rho_{\gamma^2 s/4}\, ds\right)\middle|\, \rho_{\gamma^2 u/4}, \rho_{\gamma^2 t/4}\right]\\ &= \mathbb{E}^P\left[\exp\left(-\frac{4a}{\gamma^2}\int_{\gamma^2 u/4}^{\gamma^2 t/4} \rho_s\, ds\right)\middle|\, \rho_{\gamma^2 u/4}, \rho_{\gamma^2 t/4}\right].\end{aligned}$$

Girsanov theorem

- For a fixed T, consider a probability space $(\Omega, \mathcal{F}, P)$ that accommodates a Brownian motion W_t^P, $0 \le t \le T$. Let Y_t be adapted and satisfy Novikov's condition $\mathrm{E}\left[\exp\left(\frac{1}{2}\int_0^T Y_s^2\, ds\right)\right] < \infty$.
- Define
$$X_t^{P\to Q} = \int_0^t Y_s\, dW_s^P, \qquad Z_t^{P\to Q} = \exp\left(X_t^{P\to Q} - \frac{1}{2}\int_0^t Y_s^2\, ds\right).$$
- Then
$$W_t^Q = W_t^P - \int_0^t Y_s\, ds$$
is Brownian motion under the new probability measure Q, where
$$\left.\frac{dQ}{dP}\right|_{\mathcal{F}_t, 0\le t\le T} := Z_t^{P\to Q} = \exp\left(X_t^{P\to Q} - \frac{1}{2}[X^{P\to Q}]_t\right) = \exp\left(\int_0^t Y_s\, dW_s^P - \frac{1}{2}\int_0^t Y_s^2\, ds\right),$$
with $[X]_t$ denoting the quadratic variation of a stochastic process X.
- Reversely,
$$\left.\frac{dP}{dQ}\right|_{\mathcal{F}_t} := \frac{1}{Z_t^{P\to Q}}.$$

Change of probability measure (1/3)

- The time-changed process can be re-written as
$$d\rho_s = \frac{4\kappa\bar{\nu}}{\gamma^2}\,ds + 2\sqrt{\rho_s}\,d\widetilde{W}_s^{\nu} - \frac{4\kappa}{\gamma^2}\rho_s\,ds.$$
- To compute
$$\mathbb{E}^P\left[\exp\left(-\frac{4a}{\gamma^2}\int_{\gamma^2 u/4}^{\gamma^2 t/4}\rho_s\,ds\right)\Bigg|\,\rho_{\gamma^2 u/4},\rho_{\gamma^2 t/4}\right],$$
we apply the Girsanov theorem on the time interval $[U, T]$, where
$$U = \frac{\gamma^2 u}{4}, \qquad T = \frac{\gamma^2 t}{4}.$$
- By the Girsanov theorem, for $U \leq s \leq T$,
$$W_s^G = \widetilde{W}_s^{\nu} - \int_U^s \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,dy$$
is Brownian motion under the probability measure G, with
$$\left.\frac{dG}{dP}\right|_{\mathcal{F}_s} := Z_s^{P\to G} = \exp\left(\int_U^s \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,d\widetilde{W}_y^{\nu} - \frac{1}{2}\int_U^s \rho_y\left(\frac{2\kappa}{\gamma^2}\right)^2 dy\right).$$
- Under G, the process becomes a squared Bessel process:

Change of probability measure (2/3)

- Reversely,

$$\left.\frac{dP}{dG}\right|_{\mathcal{F}_s} := Z_s^{G\to P} = \frac{1}{Z_s^{P\to G}} = \exp\left(-\int_U^s \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,d\widetilde{W}_y^{\nu} + \frac{1}{2}\int_U^s \rho_y\left(\frac{2\kappa}{\gamma^2}\right)^2 dy\right)$$

$$= \exp\left(-\int_U^s \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,dW_y^G - \frac{1}{2}\int_U^s \rho_y\left(\frac{2\kappa}{\gamma^2}\right)^2 dy\right).$$

- The process ρ under G is a squared Bessel process, i.e.

$$d\rho_s = \frac{4\kappa\bar{\nu}}{\gamma^2}\,ds + 2\sqrt{\rho_s}\,dW_s^G.$$

implying

$$\mathbb{E}^Q\left[\exp\left(-a\int_u^t \nu_s\,ds\right)\,\middle|\,\nu_u,\nu_t\right] = \mathbb{E}^P\left[\exp\left(-\frac{4a}{\gamma^2}\int_{\gamma^2 u/4}^{\gamma^2 t/4}\rho_s\,ds\right)\,\middle|\,\rho_{\gamma^2 u/4},\rho_{\gamma^2 t/4},\right]$$

$$= \frac{\mathbb{E}^G\left[\exp\left(-\frac{4a}{\gamma^2}\int_U^T \rho_s\,ds\right)Z_T^{G\to P}\,\middle|\,\rho_U,\rho_T\right]}{\mathbb{E}^G\left[Z_T^{G\to P}\,\middle|\,\rho_U,\rho_T\right]}.$$

where the last equality is due to the change of probability conditional on ρ_U and ρ_T.

Change of probability measure (3/3)

- Note that by definition

$$-\int_U^T \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,dW_y^G = -\frac{\kappa}{\gamma^2}\left(\rho_T-\rho_U-\frac{4\kappa\bar{\nu}}{\gamma^2}(T-U)\right).$$

- Thus conditional on ρ_U and ρ_T, $-\int_U^T \sqrt{\rho_y}\,\frac{2\kappa}{\gamma^2}\,dW_y^G$ can be treated as a known constant, implying that

$$Z_T^{G\to P} \propto \exp\left(-\frac{1}{2}\int_U^T \rho_y\left(\frac{2\kappa}{\gamma^2}\right)^2 dy\right),$$

where $\propto$ means equality up to a constant multiplier (check the definition of $Z_T^{G\to P}$ in the previous slide).
- Finally, we can conclude that

$$\mathbb{E}^Q\left[\exp\left(-a\int_u^t \nu_s\,ds\right)\bigg|\,\nu_u,\nu_t\right] = \frac{\mathbb{E}^G\left[\exp\left(-\left(\frac{4a}{\gamma^2}+\frac{1}{2}\left(\frac{2\kappa}{\gamma^2}\right)^2\right)\int_U^T\rho_s\,ds\right)\bigg|\,\rho_U,\rho_T\right]}{\mathbb{E}^G\left[\exp\left(-\frac{1}{2}\left(\frac{2\kappa}{\gamma^2}\right)^2\int_U^T\rho_s\,ds\right)\bigg|\,\rho_U,\rho_T\right]},$$

and from the formula for the Laplace transform of the time-integration of a squared Bessel process, we can derive the explicit formula for the ch.f. of $\int_u^t \nu_s\,ds$ given ν_t and ν_u.

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